# PROJECT PROPOSAL

## What Affects Life Expectancy? A Data Analysis of WHO Country Data

Course: Python Programming (University Level)

Guide: Prof. Vatsal Shinghala


| Student Name | Enrollment No. |
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**GitHub Repository:** https://github.com/JayJOD10/life-expectancy-analysis

## 1. Project Definition

In 2015 a person born in Japan could expect to live about 84 years (83.7), while in Sierra Leone the figure was 51.0 years. We want to find out why. Our project looks at the WHO life expectancy data for 193 countries between 2000 and 2015 and checks which health, education and economic factors move together with life expectancy.

We will clean the data, compare developed and developing countries, plot the yearly trend, measure correlations, and fit simple regression lines. At the end we will build a small Tkinter window where a user can pick a country and a factor and see the graph. The work is limited to analysis and visualization, so no machine learning is involved.

**Questions we want to answer**

- How different is life expectancy in developed and developing countries?
- Has life expectancy gone up from 2000 to 2015, and is the gap between the two groups closing?
- Which factors (schooling, adult mortality, income, immunization, GDP, etc.) are most strongly linked to it?

## 2. Dataset and Use Case

| Item | Details |
|---|---|
| Dataset | Life Expectancy (WHO), Kaggle (kumarajarshi) |
| Link | https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who |
| Size | 2,938 rows and 22 columns |
| Coverage | 193 countries (34 Developed, 159 Developing), years 2000 to 2015 |
| Source | WHO Global Health Observatory, economic data from the United Nations |

The columns cover mortality (adult mortality, infant and under-five deaths, HIV/AIDS), immunization (Hepatitis B, Polio, Diphtheria, Measles), economy (GDP, population, health expenditure, income composition of resources) and social factors (schooling, BMI, alcohol, thinness). Life expectancy in years is the value we study.

Use case: the results can help students and readers see which areas of public health and education matter most, and give us a complete practice project using the tools taught in this course.

Data quality. We loaded the file with Pandas and found missing values in 14 columns. Population (652), Hepatitis B (553) and GDP (448) have the most. Even life expectancy has 10 gaps. Missing values were filled with the median of the same country, and anything still empty with the overall median.

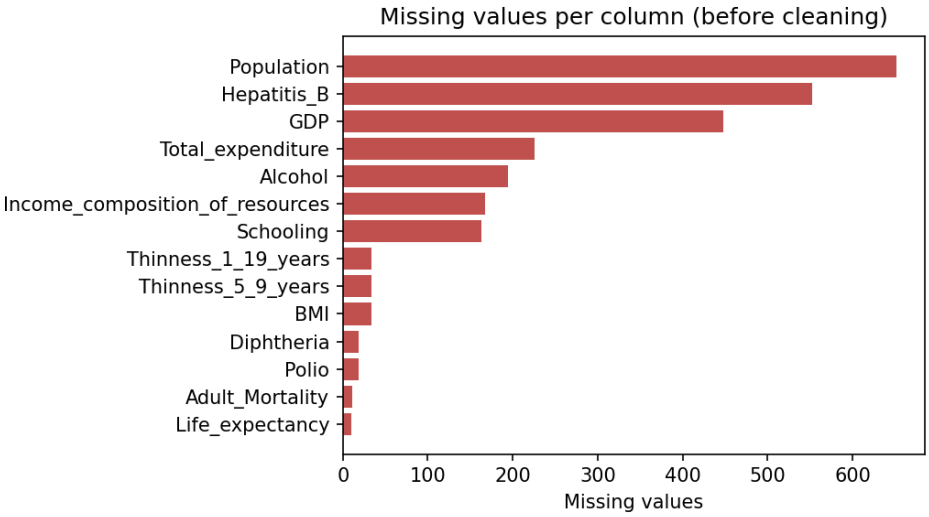



*Figure 1: Missing values per column before cleaning*

## 3. Analysis and Visualizations

We have already done a first run of the analysis on the real dataset. The graphs and numbers below come from our code (analysis.py in the repository), and the final project will extend them.

### 3.1 Distribution of life expectancy

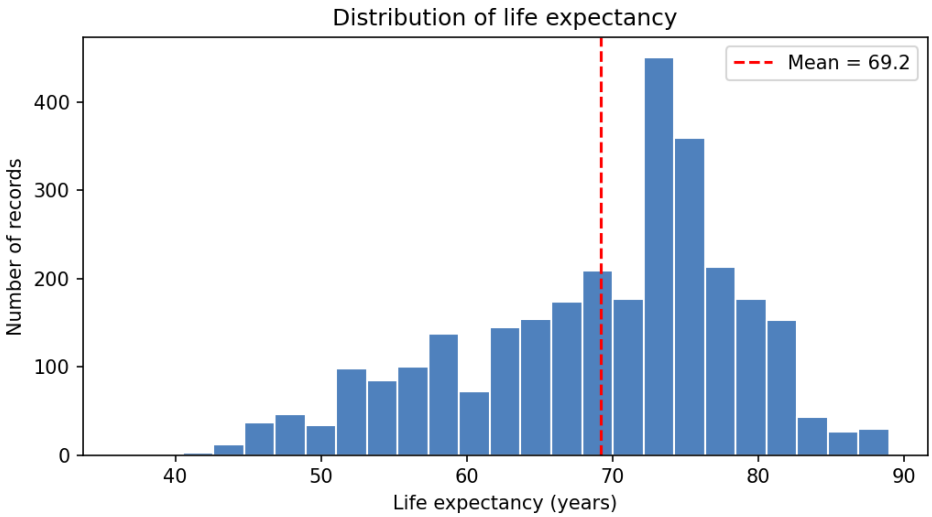



*Figure 2: Histogram of life expectancy (all countries and years)*

The average is 69.2 years and the median is 72.1. Because the mean is lower than the median, the data is skewed to the left: most records are in the 65 to 80 range, with a smaller group of countries between 36 and 55 years. The lowest value is 36.3 and the highest is 89.0.

### 3.2 Developed vs developing countries

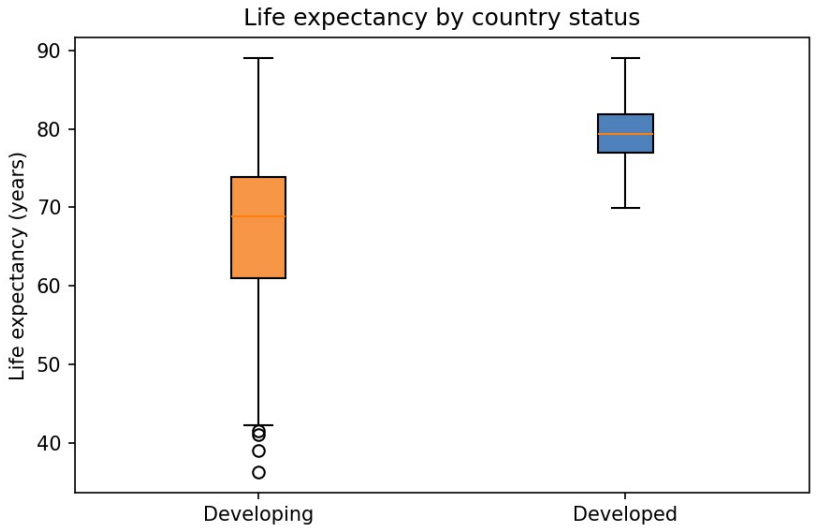



*Figure 3: Box plot of life expectancy by status*

Developed countries average 79.4 years and developing countries 66.9 years, a difference of about 12.4 years. A Welch t-test (SciPy) gives t = 50.2 and p < 0.001, so the difference is not due to chance. Developing countries also have a much wider spread, which means they vary a lot from one country to another.

### 3.3 Trend from 2000 to 2015

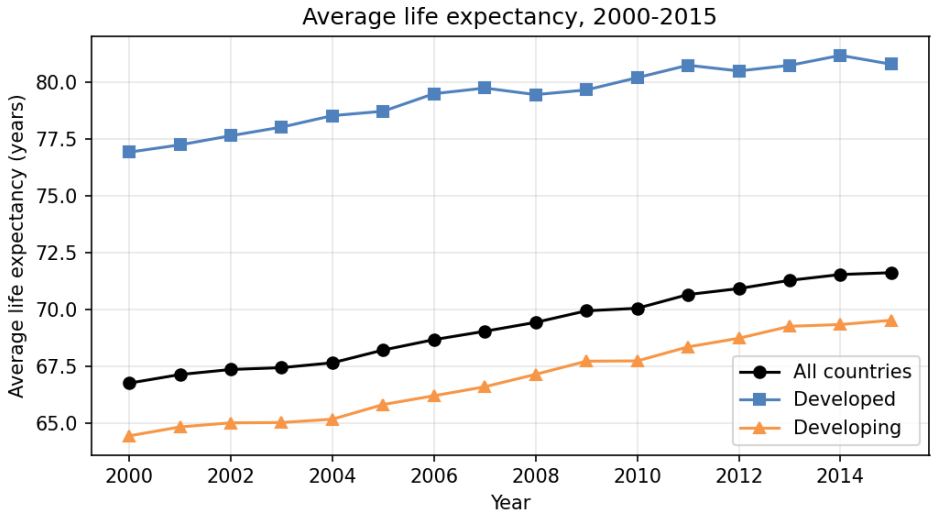



*Figure 4: Average life expectancy by year*

Average life expectancy of all countries rose from 66.8 years in 2000 to 71.6 years in 2015. Developing countries gained 5.1 years (64.4 to 69.5) while developed countries gained 3.9 years (76.9 to 80.8), so the gap between the two groups got smaller, from 12.5 to 11.3 years.

### 3.4 Correlation between factors

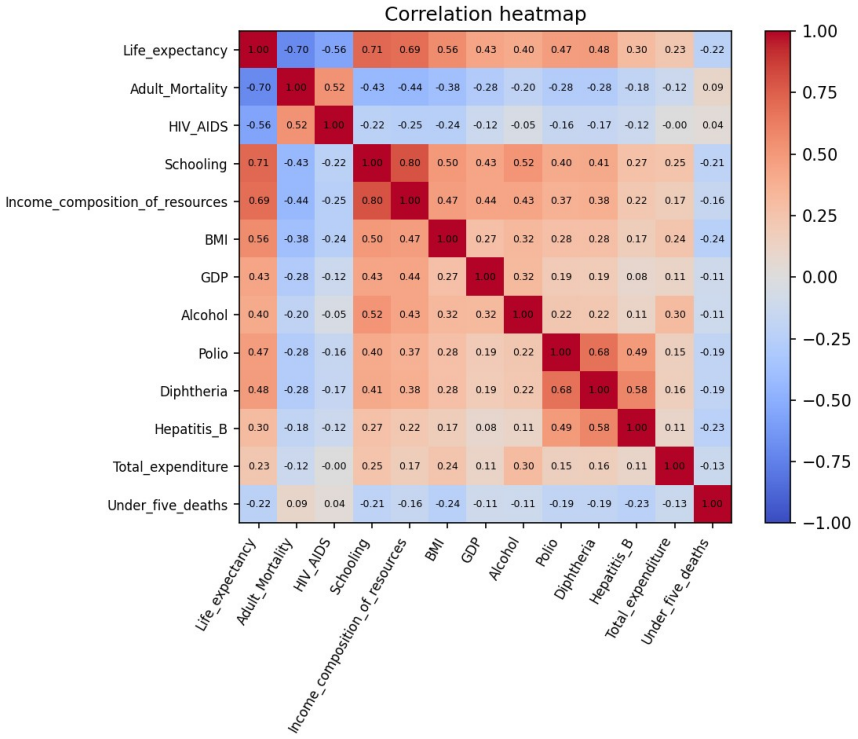



*Figure 5: Correlation heatmap of life expectancy and selected factors*

Schooling (r = 0.71) and income composition of resources (r = 0.69) have the strongest positive link with life expectancy. Adult mortality (r = -0.70) and HIV/AIDS (r = -0.56) have the strongest negative link. Polio and Diphtheria immunization are moderately positive (about 0.47 and 0.48). Schooling and income composition are also strongly related to each other (0.80), so they partly explain the same thing.

### 3.5 Scatter plots with regression lines

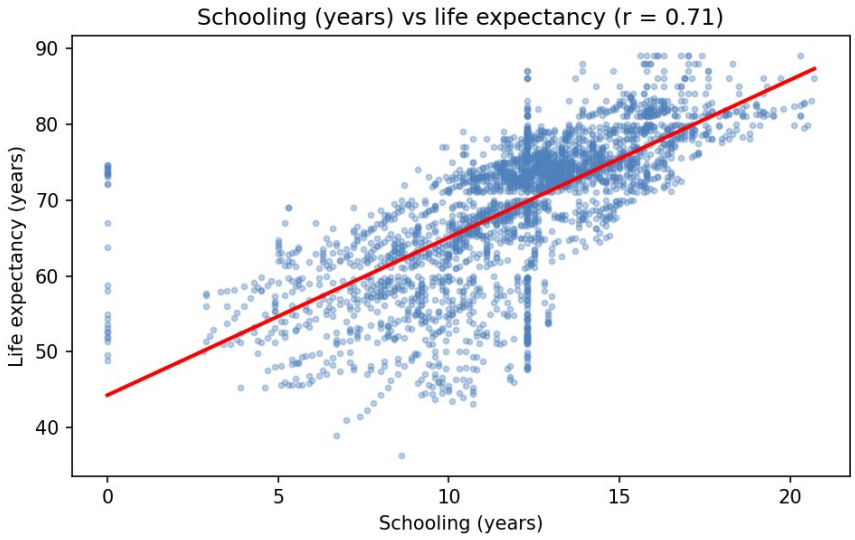



*Figure 6: Schooling vs life expectancy*

The regression line says that each extra year of schooling goes with about 2.1 more years of life expectancy (R squared = 0.51).




*Figure 7: Adult mortality vs life expectancy*

Every 100 extra adult deaths per 1000 people goes with about 5.3 fewer years of life expectancy (R squared = 0.48).

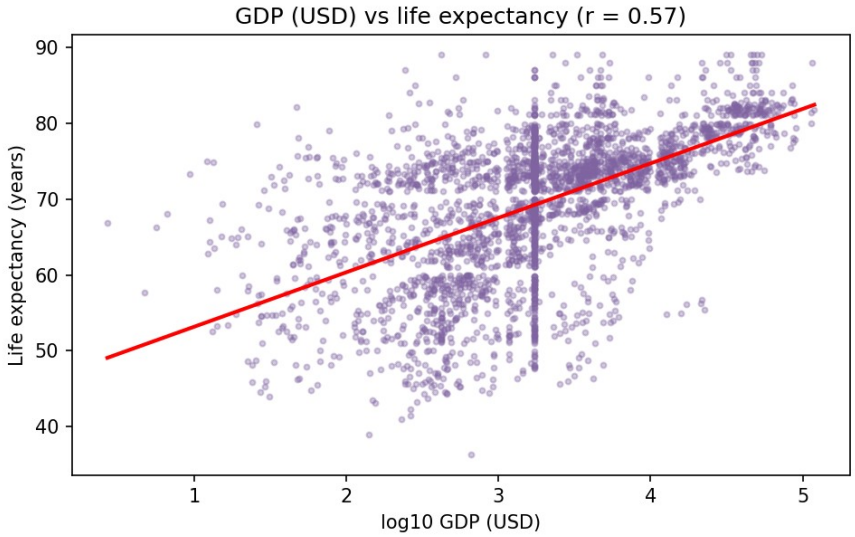



*Figure 8: GDP (log scale) vs life expectancy*

GDP needs a log scale because a few countries have very high values. With the log, the correlation is 0.57 (R squared = 0.32). Life expectancy rises quickly at low incomes and then flattens, so extra money helps less in rich countries.

### 3.6 Highest and lowest countries

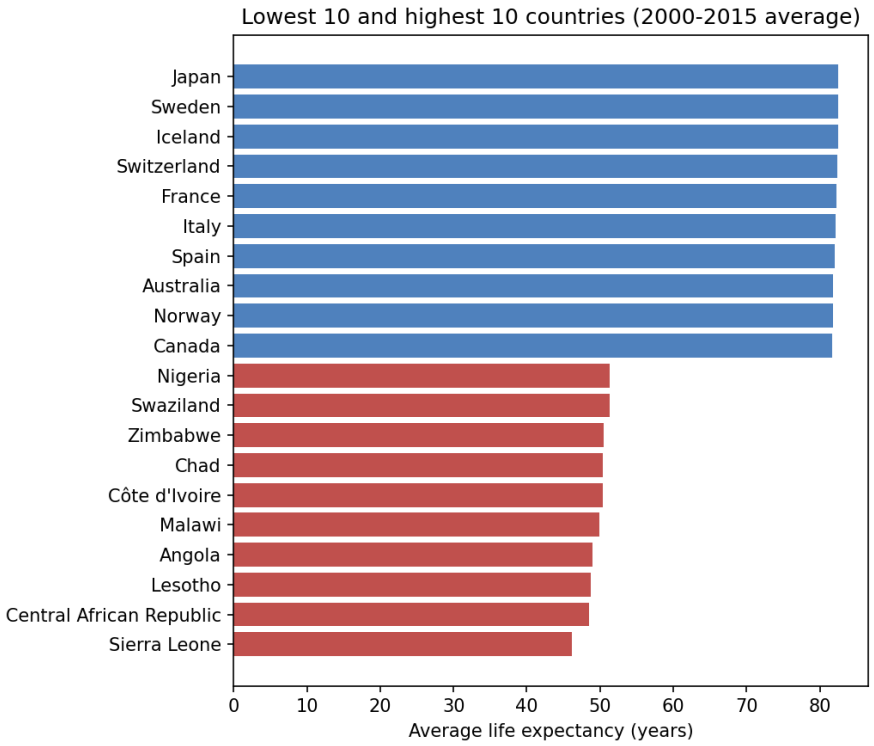



*Figure 9: Ten lowest and ten highest countries by average life expectancy*

Japan (82.5), Sweden (82.5) and Iceland (82.4) are at the top. Sierra Leone (46.1), Central African Republic (48.5) and Lesotho (48.8) are at the bottom. Most of the lowest countries are in sub-Saharan Africa.

### 3.7 Immunization

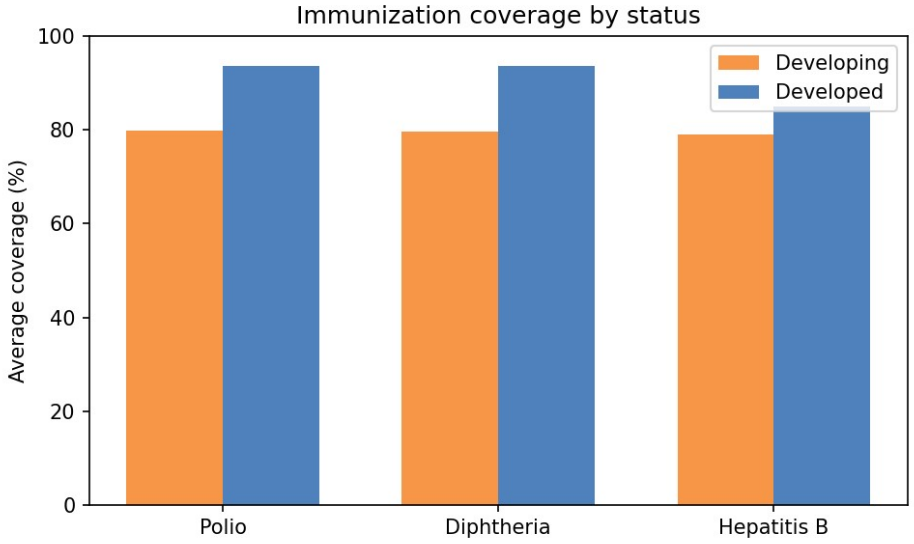



*Figure 10: Average immunization coverage by status*

Developed countries have about 94% coverage for Polio and Diphtheria against about 80% in developing countries. The gap for Hepatitis B is smaller (85% and 79%).

## 4. Summary of Statistical Results

| Test / measure | Result | Meaning |
|---|---|---|
| Mean life expectancy | 69.2 years (std 9.5) | Overall average |
| Developed vs Developing (Welch t-test) | 79.4 vs 66.9, p < 0.001 | Significant difference |
| Schooling (linear regression) | r = 0.71, slope 2.08, R2 = 0.51 | +1 year schooling, +2.1 years life |
| Adult mortality (linear regression) | r = -0.70, slope -0.053, R2 = 0.48 | Higher mortality, lower life expectancy |
| GDP, log scale (linear regression) | r = 0.57, R2 = 0.32 | Positive, flattens at high income |

A note on these results: correlation does not prove cause. Also, filling missing values with medians can slightly change the numbers, so we will compare results with and without filled values in the final report.

## 5. Python Libraries and Tools

| Library / Concept | Use in project | Syllabus unit |
|---|---|---|
| Core Python (functions, loops, dictionaries, file I/O) | Program structure, reading the CSV, helper functions | Unit I |
| OOP (classes) | Loader, Analyzer and Plotter classes | Unit II |
| re (regular expressions) | Cleaning column names | Unit II |
| NumPy | Arrays and statistics | Unit III |
| SciPy | ttest_ind, linregress | Unit III / IV |
| Matplotlib | All ten graphs | Unit III |
| Tkinter | Dashboard with country and factor dropdowns | Unit III |
| Pandas | Loading, cleaning, groupby, aggregation | Unit IV |

We did not use Seaborn or scikit-learn because they are outside our syllabus. Tools: Python 3, VS Code, Git and GitHub.

## 6. Remaining Work and Timeline

| Week | Work |
|---|---|
| Week 1-2 | First analysis and graphs (done); move the code into classes and modules |
| Week 3 | Clean column names with regular expressions; repeat the tests with and without filled values |
| Week 4 | Add more plots (country-wise trends, thinness, expenditure) |
| Week 5 | Build and test the Tkinter dashboard |
| Week 6 | Final report, README and GitHub upload |

## 7. GitHub Repository

The code (analysis.py), graphs, README and this proposal are kept in the repository linked at the top of this document.

## 8. Conclusion

Our first analysis shows a clear picture: life expectancy is about 12 years higher in developed countries, it has been rising everywhere, and the gap is slowly closing. Education and adult mortality are the factors most closely tied to it. We will now organize the code properly, finish the dashboard and write the final report.